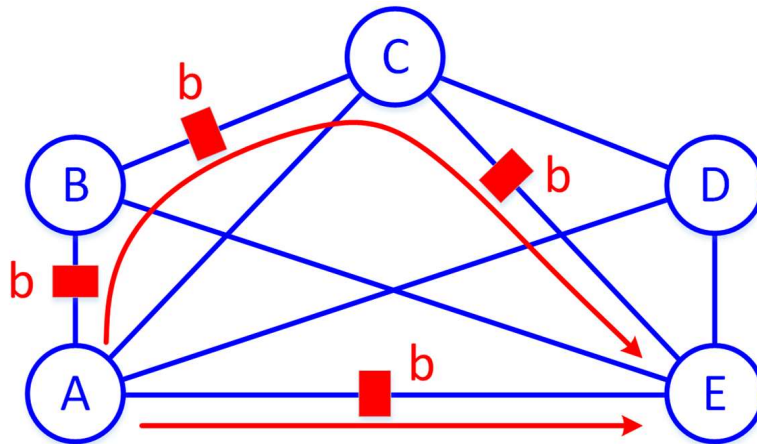


Circuit-switched networks

Circuit-switched networks

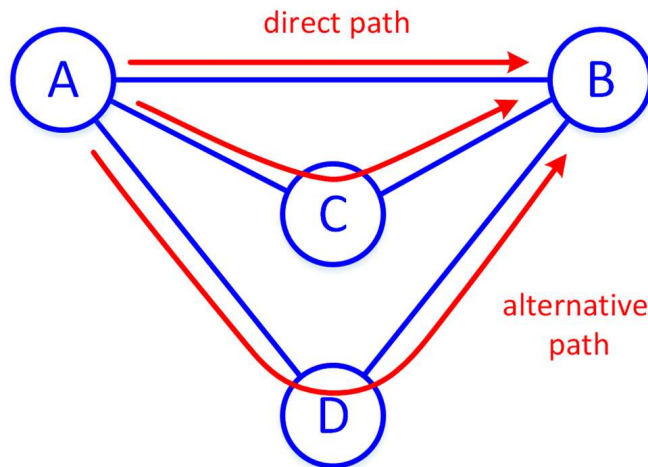
- The network processes flows of **calls** or **traffic sessions** (e.g. VoIP calls)
- Calls can be accepted or blocked – **admission control** function
- When a call of bandwidth b is accepted at the network, it occupies b bandwidth units at each link of its path



- Longer paths consume more resources
- Give preference to the **direct path** since it consumes less resource
- **Trunk reservation** - set of circuits reserved in each link for calls routed through the direct path; limits the amount of alternative routing

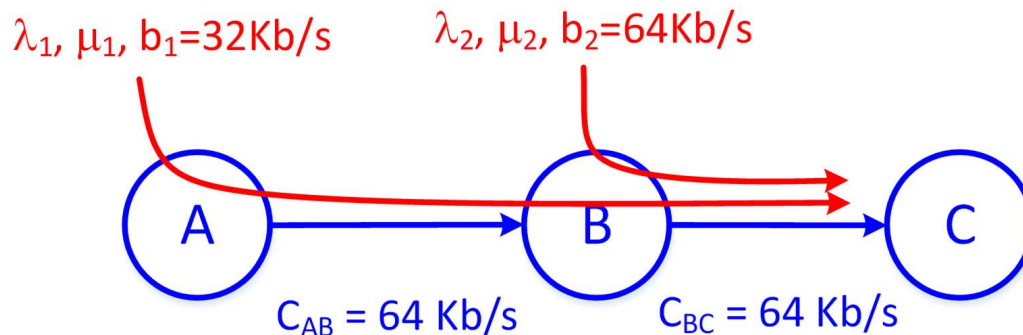
Circuit-switched networks

- For a given call flow (from some origin to some destination), routing can be:
 - **Fixed** – only a single path can be used
 - **Alternative** – a set of alternative paths can be used
 - **Dynamic** – the path (or set of alternative paths) may vary over time, depending on the congestion state of the network

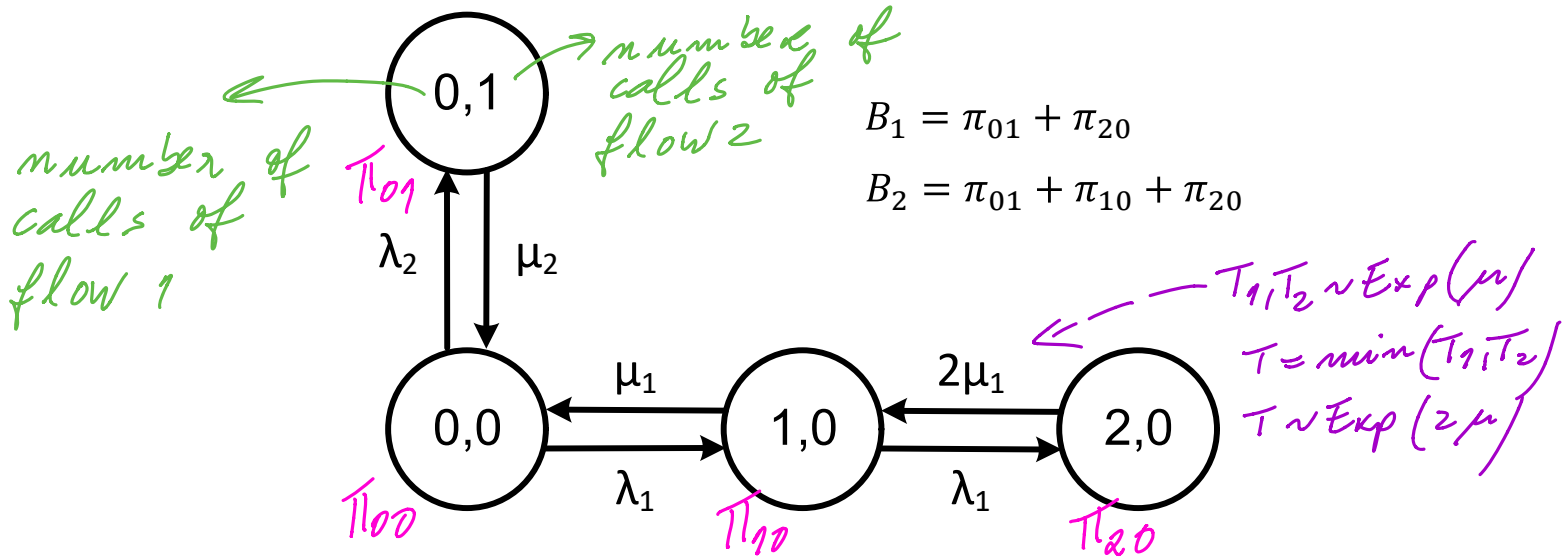
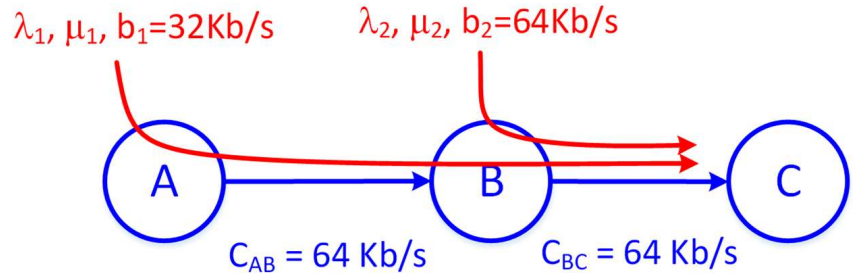


Fixed routing example

- Consider a network with 3 nodes (A, B and C) and 2 links (AB and BC). There are 2 call flows (1 and 2). Calls of flow 1 are routed through ABC and calls of flow 2 through BC. Each call of flow 1 occupies a bandwidth of $b_1 = 32$ Kb/s and each call of flow 2 a bandwidth of $b_2 = 64$ Kb/s. Both links have a capacity of $C_{AB} = C_{BC} = 64$ Kb/s. Calls arrive according to a Poisson process and the call duration is exponentially distributed. How can the blocking probability of each flow be calculated?



Fixed routing example



Multi-dimensional birth-death process that describes the system occupation

Fixed routing (general case)

- Consider a network with J links, where link j has capacity C_j , that serves K call flows;
- Each flow k has a call arrival rate, λ_k , an average call duration $1/\mu_k$ (a traffic intensity $\rho_k = \lambda_k/\mu_k$), a call bandwidth b_k and a path $R_k \subseteq \{1, 2, \dots, J\}$;
- The calls of the K flows arrive according with independent Poisson processes at rate λ_k ;
- A call of flow k that has been admitted in the network, occupies b_k bandwidth during the service time, which is exponentially distributed with mean $1/\mu_k$;
- The call durations are independent and independent from the arrival processes;

Fixed routing (general case)

- Let n_k be the number of calls of flow k in the network, $\mathbf{n} = (n_1, \dots, n_k)$, and K_j the set of flows that cross link j ;
- A call from flow k will not be admitted in the network if

$$b_k + \sum_{l \in K_j} b_l n_l > C_j$$

in at least one link belonging to the path of k ;

*bandwidth
occupied at link j
by calls that are
being carried out*

Fixed routing (general case)

- The state space of the multi-dimensional birth-death process is

$$S = \left\{ \mathbf{n} \in \mathbf{I}^k: \sum_{k \in K_j} b_k n_k \leq C_j, \quad j = 1, \dots, J \right\}$$

where $\mathbf{I}$ is the set on non-negative integers;

- Let S_k be the subset of states for which a call of flow k is accepted when is it offered to the network, i.e.

$$S_k = \left\{ \mathbf{n} \in S: \sum_{l \in K_j} b_l n_l \leq C_j - b_k, \quad j \in R_k \right\}$$

Fixed routing (general case)

- The state limiting probabilities are given by

$$\pi(\mathbf{n}) = \frac{1}{G} \prod_{k=1}^K \frac{\rho_k^{n_k}}{n_k!}, \quad \mathbf{n} \in S$$

where

$$G = \sum_{\mathbf{n} \in S} \prod_{k=1}^K \frac{\rho_k^{n_k}}{n_k!}$$

Fixed routing (general case)

- The blocking probability of flow k is given by

$$B_k = 1 - \frac{\sum_{\mathbf{n} \in S_k} \prod_{l=1}^K \frac{\rho_l^{n_l}}{n_l!}}{\sum_{\mathbf{n} \in S} \prod_{l=1}^K \frac{\rho_l^{n_l}}{n_l!}}$$

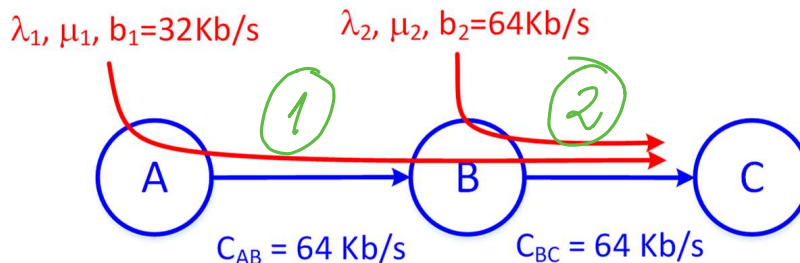
Fixed routing (general case)

$$B_k = 1 - \frac{\sum_{n \in S_k} \prod_{l=1}^K \frac{\rho_l^{n_l}}{n_l!}}{\sum_{n \in S} \prod_{l=1}^K \frac{\rho_l^{n_l}}{n_l!}}$$

- $J = 2, K = 2$
- Flow 1: $\lambda_1, \mu_1, b_1, R_1 = \{1, 2\}$; Flow 2: $\lambda_2, \mu_2, b_2, R_2 = \{2\}$
- $S = \{(0,0), (1,0), (2,0), (0,1)\}$, $S_1 = \{(0,0), (1,0)\}$, $S_2 = \{(0,0)\}$

$$B_1 = 1 - \frac{\frac{\rho_1^0 \rho_2^0}{0! 0!} + \frac{\rho_1^1 \rho_2^0}{1! 0!}}{\frac{\rho_1^0 \rho_2^0}{0! 0!} + \frac{\rho_1^1 \rho_2^0}{1! 0!} + \frac{\rho_1^2 \rho_2^0}{2! 0!} + \frac{\rho_1^0 \rho_2^1}{0! 1!}}$$

$\underbrace{\quad}_{n_1=0, n_2=0} \quad \underbrace{\quad}_{n_1=1, n_2=0} \quad \underbrace{\quad}_{n_1=2, n_2=0} \quad \underbrace{\quad}_{n_1=0, n_2=1}$



Product bound

- An approximation to calculate the flow blocking probabilities.
- Applies only when the calls of all flows require the same bandwidth (single-service networks).
- Assumes that blocking is independent from link to link and that load applied to link is the sum of loads over all flows that cross the link.
- It gives an upper-bound of the flow blocking probability.
- It is a good approximation when:
 - The flows cross a small number of links
 - The blocking probabilities are small

Product bound

- Let

$$\bar{\rho}_j = \sum_{k \in K_j} \rho_k$$

$$ER(\rho, K) = \frac{\rho^K / K!}{\sum_{i=0}^K \rho^i / i!}$$

→ Erlang B formula

- Then, the blocking probability of flow k is

$$B_k \leq 1 - \prod_{j \in R_k} (1 - ER(\bar{\rho}_j, C_j))$$

Probability of call being accepted at all links of its path

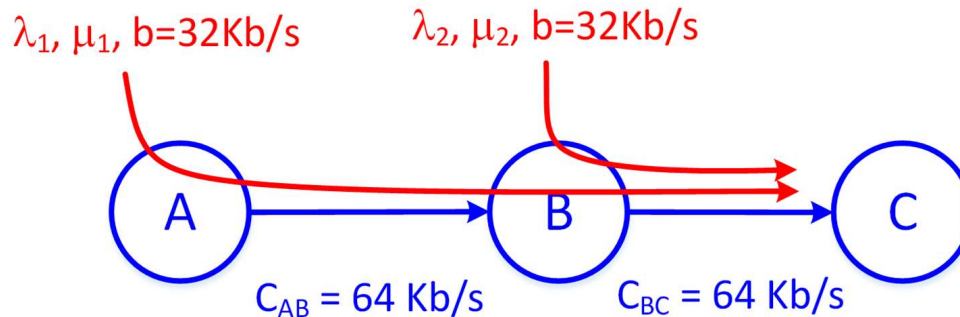
where C_j is the capacity expressed as number of simultaneous calls

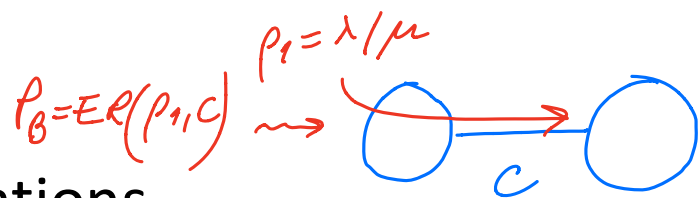
Product bound example

- $L_{AB} = ER(\rho_1, 2)$
- $L_{BC} = ER(\rho_1 + \rho_2, 2)$
- $B_1 \leq 1 - (1 - L_{AB})(1 - L_{BC})$
- $B_2 \leq 1 - (1 - L_{BC}) = L_{BC}$

Approximations:

- 1- Acceptance of call is independent from link to link*
- 2- Traffic intensity offered to link is the sum of the flow traffic intensities*





Product bound limitations

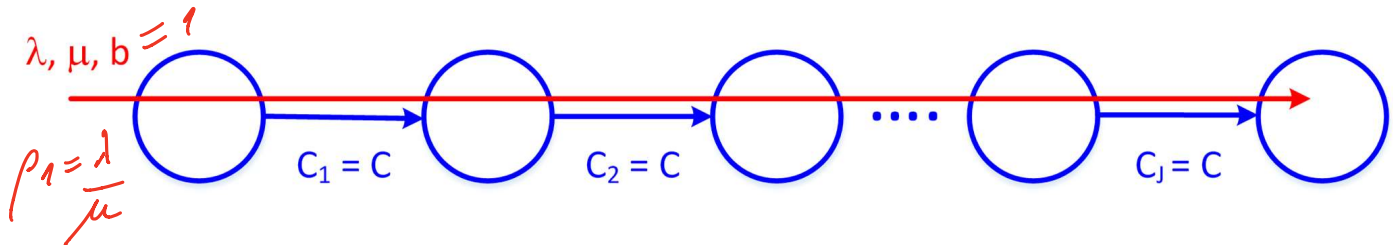
linear network

(expressed in number of circuits)

- Consider a ~~path~~ with J links, each one with capacity C , and a single call flow routed through all links. In this case,

$$B_1 \leq 1 - (1 - ER(\rho_1, C))^J \quad \begin{array}{l} \longrightarrow 1 \\ J \rightarrow \infty \end{array}$$

- The upper bound goes to 1 when J goes to infinity. However, the exact blocking probability is $ER(\rho_1, C)$, for any J . This shows that this approximation can be too coarse when calls are routed through many links.



Reduced load approximation

- Idea: to reduce the traffic offered to link j taking into account the blocking in the remaining links of the path.
- Considering that the blocking probability of link j is

$$L_j = ER \left(\sum_{k \in K_j} \rho_k t_k(j), C_j \right)$$

where $t_k(j)$ is the probability of having at least one circuit available in each link of $R_k - \{j\}$.

Reduced load approximation

- Considering that call blocking is independent from link to link, it results:

$$t_k(j) = \prod_{i \in R_k - \{j\}} (1 - L_i)$$

- Combining the two previous results:

$$L_j = ER \left(\sum_{k \in K_j} \rho_k \prod_{i \in R_k - \{j\}} (1 - L_i), C_j \right), \quad j = 1, 2, \dots, J$$

- These equations are called **fixed point equations**.

Reduced load approximation

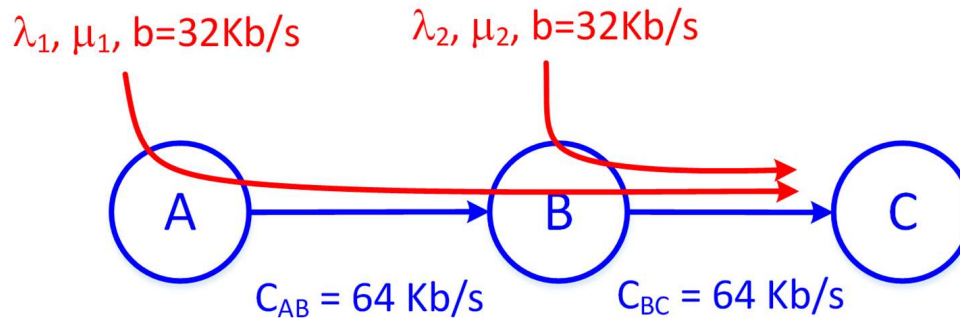
- Assuming again that the blocking probability is independent from link to link, it results for the blocking probability of flow k

$$B_k \approx 1 - \prod_{j \in R_k} (1 - L_j), \quad k = 1, 2, \dots, K$$

- It can be shown that the reduced load approximation has a unique solution.

Reduced load example

- $L_{AB} = ER(\rho_1(1 - L_{BC}), 2)$
- $L_{BC} = ER(\rho_1(1 - L_{AB}) + \rho_2, 2)$
- $B_1 \approx 1 - (1 - L_{AB})(1 - L_{BC})$
- $B_2 \approx 1 - (1 - L_{BC}) = L_{BC}$



Reduced load approximation

- The fixed point equations can be solved by **repeated substitutions**.

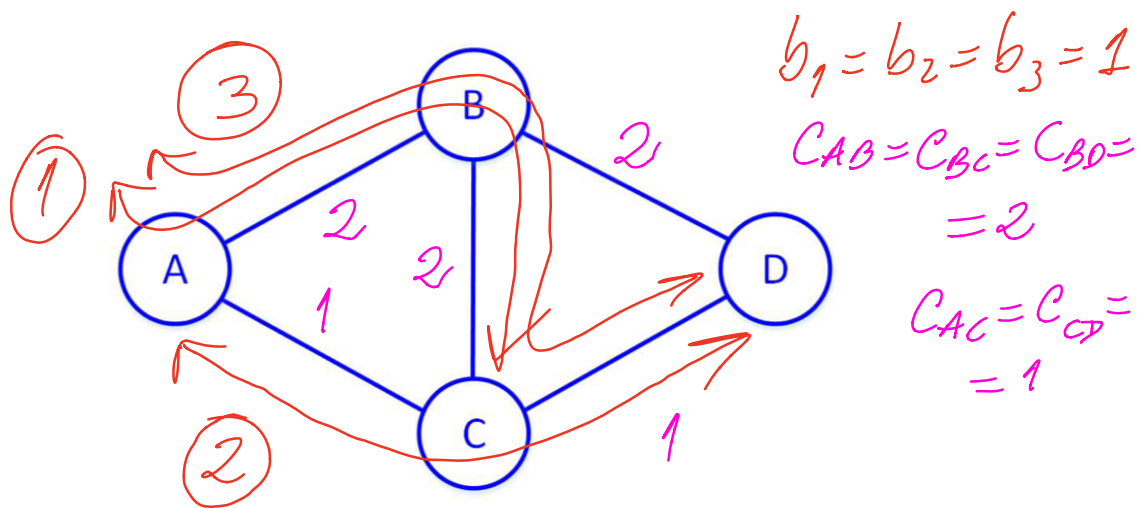
- Let $\mathbf{L} = (L_1, L_2, \dots, L_J)$ and $\mathbf{T}(\mathbf{L}) = (T_1(\mathbf{L}), T_2(\mathbf{L}), \dots, T_J(\mathbf{L}))$, where

$$T_j(\mathbf{L}) = ER \left(\sum_{k \in K_j} \rho_k \prod_{i \in R_k - \{j\}} (1 - L_i), C_j \right)$$

- Start with some $\mathbf{L}$ (e.g. all zeros) and then iteratively apply the operator $\mathbf{T}$ for n iterations:

$$\begin{aligned} \mathbf{L}^0 &= \mathbf{L} \\ \mathbf{L}^m &= \mathbf{T}(\mathbf{L}^{m-1}), \quad m = 1, 2, \dots, n \end{aligned}$$

- Convergence is not guaranteed, but happens for most networks of practical interest.



Product bound:

$$L_{AB} = ER(p_1 + p_3, 2)$$

$$L_{AC} = ER(p_2, 1)$$

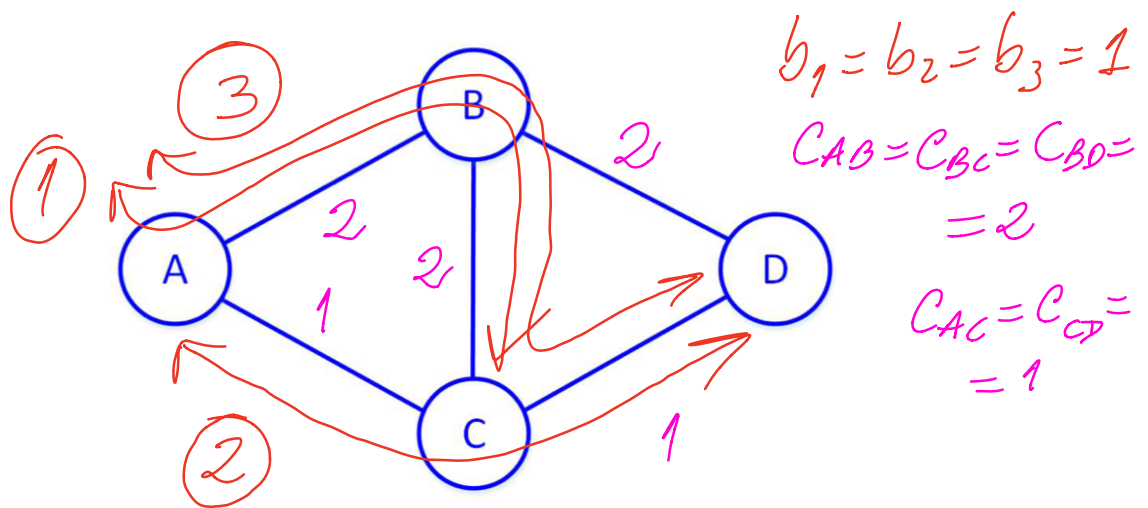
$$L_{BC} = ER(p_1 + p_3, 2)$$

$$L_{CD} = ER(p_2 + p_3, 1)$$

$$B_1 \leq 1 - (1 - L_{AB})(1 - L_{BC})$$

$$B_2 \leq 1 - (1 - L_{AC})(1 - L_{CD})$$

$$B_3 \leq 1 - (1 - L_{AB})(1 - L_{BC})(1 - L_{CD})$$



Reduced load

$$L_{AB} = ER(\rho_1(1-L_{BC}) + \rho_3(1-L_{BC})(1-L_{CD}), 2)$$

$$L_{AC} = ER(\rho_2(1-L_{CD}), 1)$$

$$L_{BC} = ER(\rho_1(1-L_{AB}) + \rho_3(1-L_{AB})(1-L_{CD}), 2)$$

$$L_{CD} = ER(\rho_2(1-L_{AC}) + \rho_3(1-L_{AB})(1-L_{BC}), 1)$$

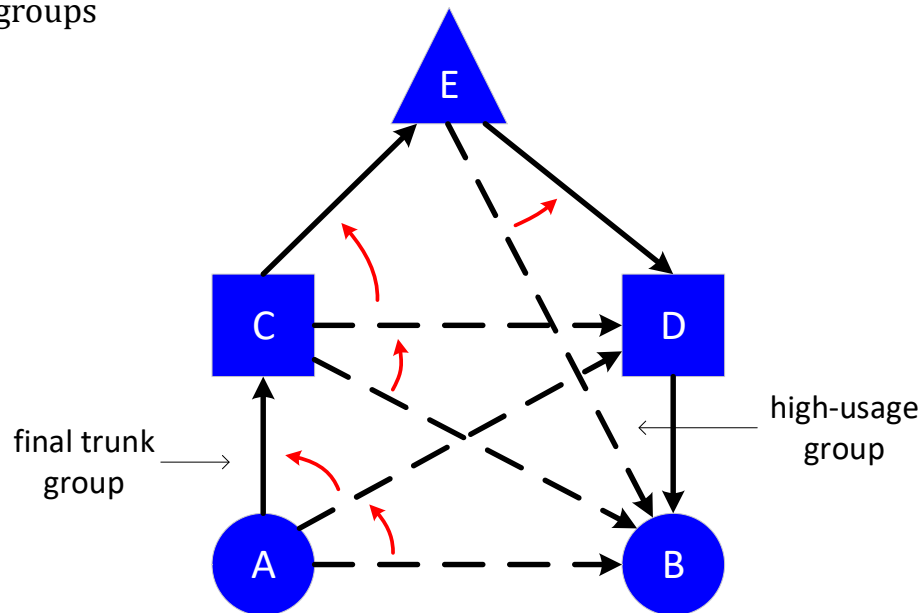
$$B_1 \approx 1 - (1-L_{AB})(1-L_{BC})$$

$$B_2 \approx 1 - (1-L_{AC})(1-L_{CD})$$

$$B_3 \approx 1 - (1-L_{AB})(1-L_{BC})(1-L_{CD})$$

Hierarchical routing in the telephony network

- Alternative routing where telephony exchanges are organized in levels.
 - Oldest routing method in circuit-switched networks
 - Subscribers are directly connected to end offices (local exchanges); each exchange is connected to the exchange of the next upper-level through a final trunk group; an exchange can be connected to other exchanges through high-usage groups



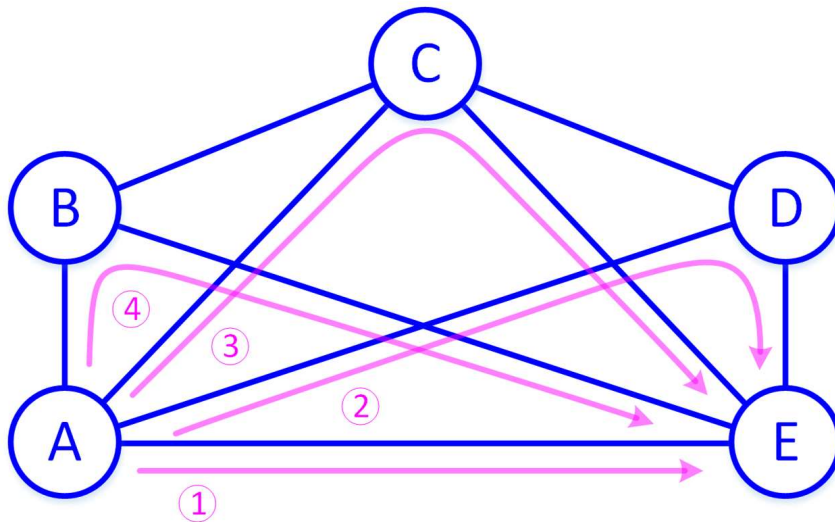
Dynamic routing

- Route first through the direct path; if not available try the available alternative routes (usually only paths with two links are used)
- Methods:
 - Sequential – try ordered subset of alternative paths, one subset for each OD pair; the subsets can change over time; used by AT&T (DNHR)
 - Random – select one path at random from a list of alternative paths; if path is full call is blocked
 - Sticky random routing – only one alternative path available at a time; if call is blocked, select another alternative path, randomly, to be used on subsequent calls; used by British Telecom (DAR)
 - Least loaded routing – try the path[^] with the most idle capacity; used by AT&T (RTNR)

first

DNHR (Dynamic Non-Hierarchical Routing)

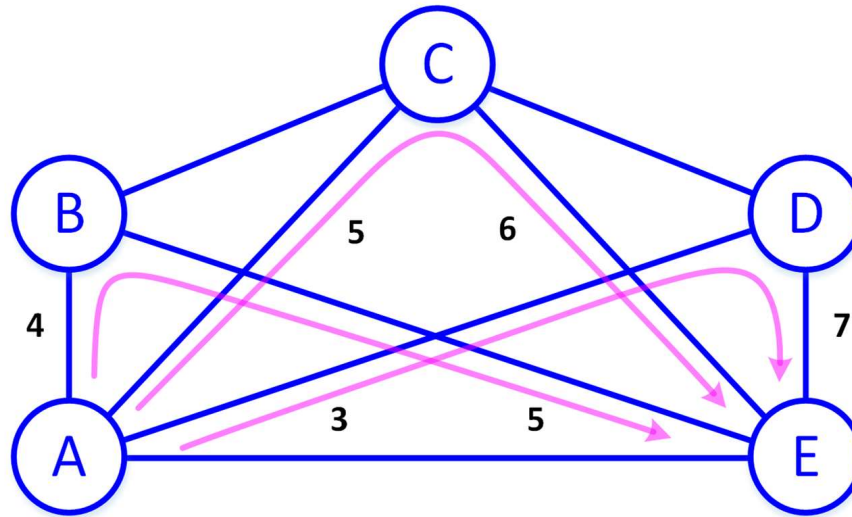
- Sequential routing method introduced by AT&T in the 80's; each day is split in 10 consecutive times periods



Time period	Path sequence
10:00-12:00	1→3→2→4
12:00-17:00	1→4→2
17:00-19:00	1→4→3→2
19:00-23:00	1→2→3
23:00-10:00	1→2

RTNR (Real-Time Network Routing)

- Least loaded routing method introduced by AT&T in the early 90's; relatively complex in terms of signaling.



- Figure shows available circuits in each link (of A to E 2-link routes).
- The available capacity of a path is the minimum among its links.
- Which is the selected route?